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31 July 2026

To the Editors, *Physical Review D*

I submit for your consideration a short article, *"The MOND acceleration scale is one half of the dark-energy gravitational rate."*

Context. The acceleration scale a_0 that organizes galactic rotation curves has been known since Milgrom (1983) to lie near cH_0, and the coincidence has attracted a series of proposed closed forms — a derived $2cH_\Lambda$ from a de Sitter–Unruh argument (Milgrom 1999; also Pikhitsa 2010, Klinkhamer & Kopp 2011), and an empirical $cH_\Lambda/2\pi$ (Milgrom 2020). This manuscript records an exact expression in the dark-energy density with a rational coefficient, and is careful to establish what that does and does not add to the above.

Key findings.

- $a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda} = c^2\sqrt{\Lambda/32\pi} = cH_\Lambda/\sqrt{32\pi/3}$. The three forms are algebraically identical, and the reduction cancels every factor of π and the 3, which shows that the apparent structure $\sqrt{32\pi/3}$ is only a unit conversion and carries no content.
- On Planck 2018 inputs the expression gives $9.43\times10^{-11}\,\mathrm{m\,s^{-2}}$, which is 0.70% from the value obtained by fitting the 175 SPARC rotation curves at $\Upsilon_{[3.6]} = 0.70$ — a mass-to-light ratio inside the range expected for Spitzer $3.6\,\mu$m populations, so the agreement does not require an implausible fit.
- The relation carries a falsifiable consequence (Sec. VI): if $a_0 \propto \sqrt{\rho_\Lambda}$ then $a_0(z)$ is exactly constant for $w = -1$ and rises to a maximum before *declining* for the DESI-preferred $w_0 > -1$, $w_a < 0$, in contrast to the monotonic rise predicted by $a_0 \propto cH(z)$. A measured monotonic rise would exclude the relation.

What I have not claimed, stated here because it bounds the paper. The interpolating function I use is *identical* to Eq. (9) of Milgrom (1999) — not a variant of it — and that same paper *derives* $a_0 = 2cH_\Lambda$, a factor 11.58 larger. My contribution is therefore a renormalization of his coefficient to match data, not a new derivation, and Sec. II says so in those words. Further, the coefficient here is not observationally distinguishable from Milgrom's $cH_\Lambda/2\pi$: the two differ by 7.9%, which is 0.018 dex in $g_{\rm obs}$ against a fit scatter of 0.108 dex. I make no claim of improved fit; the contribution is one of *form.* Section V states that the coefficient $\frac{1}{2}$ is fitted, not derived, and reports my own failed attempt to force it. Section VII hands over the outstanding tensions unprompted, including that the cluster-scale discrepancy is 13% *worse* for this lower a_0 than for the conventional value, and that the exact interpolation leaves a solar-system residual three orders above Earth–Mars ranging bounds.

Reproducibility and disclosure. Every numerical claim is produced by a public script that performs internal consistency checks and exits with a nonzero status if any fails; the code is archived at `https://doi.org/10.5281/zenodo.21724575`. Section VIII discloses that portions of the analysis and drafting were carried out with AI assistance: I directed the work, specified and reviewed every load-bearing calculation, and take full responsibility including for any errors.

No AI system is listed as an author. Several intermediate claims produced during the work were found to be incorrect and were withdrawn before submission.

Submission history. No version of this manuscript has previously been submitted to *Physical Review D* or to any other *Physical Review* journal, nor to any other journal. It is not part of a joint submission or a sequence of papers. A preprint is deposited at `https://doi.org/10.5281/zenodo.21724575`, concurrently with this submission; it is not a journal publication. The manuscript is not posted to arXiv at the time of submission, and I will inform the editorial office if that changes.

Recommended referees. I would welcome review by researchers familiar with the acceleration-scale literature, since the value of this manuscript rests largely on whether its treatment of prior work is judged correct:

- Stacy S. McGaugh, Case Western Reserve University
- Benoît Famaey, Observatoire astronomique de Strasbourg
- Federico Lelli, INAF — Osservatorio Astrofisico di Arcetri
- Harry Desmond, Institute of Cosmology and Gravitation, University of Portsmouth

Excluded referees. None.

I am an independent researcher and have no funding to declare. I am content for the manuscript to be assessed as a short paper.

With thanks for your time,

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